

DRY-RUN GUIDE | temp_convert.js

Source: javascript\1st_belt\temp_convert.js

CORE BEHAVIOR

convertTemperature returns a two-decimal Celsius/Fahrenheit conversion; top-level code reads temperature and unit from two input lines and prints the returned string.

REPRESENTATIVE INPUT / CALL

Input '32' then 'F'.

STEP TRACE

F branch computes $(32-32)*5/9=0$; toFixed(2) yields '0.00'.

EXPECTED OUTPUT / RETURN

'0.00'

ACTUAL SOURCE EXCERPT

Verbatim lines from the source; long lines wrap. Source files were not modified.

```
001 | // Question: Convert temperature from Celsius to Fahrenheit.
002 | // Add your solution here.
003 |
004 | const fs = require('fs');
005 |
006 | function convertTemperature(temp, unit) {
007 |     if (unit === 'C') {
008 |         return ((temp * 9/5) + 32).toFixed(2);
009 |     } else {
010 |         return ((temp - 32) * 5/9).toFixed(2);
011 |     }
012 | }
013 |
014 | // Read all inputs from standard input
015 | const input = fs.readFileSync(0, 'utf-8');
016 |
017 | // Split the input by lines and remove any leading/trailing whitespace
018 | const lines = input.trim().split('\n');
019 |
020 | // Parse temperature as a floating-point number and get the unit
021 | const temp = parseFloat(lines[0].trim());
022 | const unit = lines[1].trim();
023 |
024 | // Perform the conversion and print the output
025 | const result = convertTemperature(temp, unit);
026 | console.log(result);
027 |
```